Parva Barot

Tempe Area, AZ | +1 (602) 565-7895 | pbarot3@asu.edu | Linkedin

EDUCATION :

Arizona State University **[August,2023 – May,2025]**

Master's in Information Technology (GPA: 4.00/4.00)

Pandit Deendayal Energy University **[August 2019 – May 2023]**

Bachelor of Technology in Computer Science and Engineering (GPA: 3.53/4.00)

TECHNICAL SKILLS:

- **Programming Languages:** JavaScript, C, C++, Python, SQL
- **Web Development:** PHP, Redux, HTML, CSS, React, Django
- **Technologies:** Data Structures, Data Analysis, Data Science, Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Visualization, Artificial Intelligence, Big Data Analytics, Statistical Data Analysis
- **Cloud:** AWS Services(Lambda, S3, Dynamo DB, MQTT, SES, SNS, QuickSight, CloudWatch), AWS IOT, Cloud Computing
- **Databases:** MySQL, DBMS
- **Tools:** Power BI, Tableau, VS Code, Anaconda, Thonny, Raspberry Pi
- **Others:** Leadership, Teamwork, Problem-Solving, Project Management, Scrum, Agile Methodologies, Risk Crisis and Disaster Management

PROFESSIONAL EXPERIENCE:

Software Development Intern, Tntra, Ahmedabad, [Movie Revenue Prediction] **[January,2023 - April,2023]**

- Built a machine learning model using Python and linear regression to forecast movie revenue based on factors such as budget and release year.
- Implemented data preprocessing methods to address missing data and derive essential features, enhancing the model's predictive accuracy.
- Assessed model performance using Mean Squared Error and R-squared, delivering valuable insights for stakeholders in the entertainment sector.

React Developer Intern, Tntra, Ahmedabad **[May,2022 – June,2022]**

- Led a React Project on a Video Streaming and Sharing platform, employing React-Redux, React-Hooks, and API for calling the user's data.
- Employed JavaScript, APIs, and Redux, for Video streaming and Sharing Platforms.
- Develop reusable components and front-end libraries for future use.

Software Development Intern, Disha Enterprise, Ahmedabad, [Loan Defaults Prediction] **[May,2021 - July, 2021]**

- Built a machine learning model (Random Forest, Logistic Regression) to predict loan defaults, improving accuracy by 15%.
- Handled missing values, scaled features, and addressed class imbalance using oversampling and class weights.
- Evaluated model using accuracy, precision, recall, and AUC-ROC, effectively identifying high-risk loan applicants.

PROJECTS:

Smart Lighting System : Raspberry Pi, LDR, PIR, MQTT, AWS services **[January,2025- May,2025]**

- Designed and implemented a real-time Smart Lighting System using Raspberry Pi, LDR, and PIR sensors with dynamic LED control based on ambient light and motion detection logic.
- Developed a two-way communication system using AWS IoT Core and MQTT protocol to remotely trigger sensor readings and manual LED controls (ON/OFF) via Lambda functions.
- Integrated AWS services (IoT Core, Lambda, CloudWatch Logs, and SNS) to log sensor data, enable remote command execution, and trigger email notifications for key events, achieving scalable and cloud-controlled automation.

Airbnb New York – Data Visualization (Tableau): Interactive Dashboard **[June,2024- August,2024]**

- Analyzed Airbnb NYC data and created an interactive dashboard to display trends in pricing, cancellations, and reservations.
- Presented neighborhood-specific insights to aid decision-making for key stakeholders, including tourists, hosts, and local authorities.
- Provided data-driven recommendations to enhance understanding of market conditions and improve strategic planning.

Grammar Autocorrector System: RNN, T5, BERT **[January,2024 - May,2024]**

- Developed a Python-based system utilizing transformer NLP models to identify and correct grammatical errors in text.
- Applied pre-trained models like T5 and BERT for sequence-to-sequence tasks, achieving accurate, context-based corrections.
- Implemented the system with a user-friendly interface, enabling real-time text correction for content creation applications.

Mushroom Cultivation in an Agile Environment: **[January,2023- July,2023]**

- Built an Automated Mushroom Cultivation tent, and humidifier, which included Raspberry PI, and Asus Tinker Board, to manage and control the environment inside the tent.
- Crafted using Python, and a web application to view and manage the humidity and temperature inside the tent.
- Used Blynk cloud to store the data received from the sensors for better analysis.

Gesture Control: Virtual Mouse for Contact-Free Interaction: CNN, OpenCV, Python **[September,2022- December,2022]**

- Developed a contact-free gesture control system using Python, OpenCV, and MediaPipe for real-time hand gesture recognition.
- Implemented hand pose estimation and finger tracking algorithms to map gestures to mouse movements and actions.
- Designed a virtual mouse interface to allow intuitive, touchless interaction with computers, enhancing accessibility and user experience.

COURSE WORK:

Certification course in The Joy of Computing Using Python in NPTEL, Udemy Certification course in Ethical Hacking. Udemy Certification course in Android App Development, Udemy Certification course in Machine Learning, Tata Imagination Challenge 2022, **Global Immersion Program & Curtin University Dubai**: For develop skills in Project Management.

PUBLICATIONS:

Mushroom Cultivation in an Agile Environment:

Kathiria, A., Barot, P., Paliwal, M., Shastri, A. (2024). IoT-Assisted Mushroom Cultivation in Agile Environment. In: Tiwari, S., Trivedi, M.C., Kolhe, M.L., Singh, B.K. (eds) Advances in Data and Information Sciences. ICDIS 2023. Lecture Notes in Networks and Systems, vol 796. Springer, Singapore. DOI: https://doi.org/10.1007/978-981-99-6906- URL: https://link.springer.com/chapter/10.1007/978-981-99-6906-7_26#citeas